

Presentation of a paper

Avoiding Invalid Instruments and Coping with Weak Instruments
Murray, M. P. (2006).

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Background of the author

Academic and non academic

Michael P. Murray main research interest in poverty, with articles on economic theory and econometrics.

Taught urban, public, development, behavioral economics, microeconomics theory and econometrics. At Bates University, visiting professorship at UC Berkley, Harvard, LUISS Guido Carli University.

Non academic career includes positions at RAND Corporation & the World Bank.

Defining the main concern in IV

Invalid and Weak Instruments

Instrument Variable estimation is used to avoid biased results that arise when independent variables are correlated with the error term. This issue is addressed using a two-stage estimation (2SLS) or multi stage estimation process, where the first stage predicts the problematic variable with instruments, and the second or the additional stages uses these predictions to estimate the outcome variable.

Invalid Instrument Variable are IVs that are correlated with the error term.

Weak Instrument Variable are those that are weakly correlated with the independent variable, and thus do not effectively correct for biases.

Practical Analogy

Archimedes' Lever and the Earth



Type of IV = Lever Material

Valid = Titanium

Invalid = Bamboo / Rubber

Weak = Plastic / Wood

Invalid IVs

Ensuring validity of Instrumental Variables (IVs)

Relevance

Instruments must be correlated with the endogenous explanatory variable (incarceration rates, police numbers.)

Exclusion Restriction

instruments should affect the dependent variable (e.g crime rate) only through the endogenous variable.

Preclusion of Links to Disturbances

verify empirical checks (lawsuits predicting past crime rates) that instruments are not related to unobserved shocks affecting the outcome.

Invalid IVs

Strategies to Validate Instruments and Avoid Bias

Test over-identifying restrictions

Use tests like Sargan's to check if instruments are consistent with the data.

$$S = n * R^2$$

Use Multiple Instruments

Comparing results from different instruments (e.g., election cycles vs. firefighters) enhances credibility; consistent estimates support validity.

Diligent Omitted Variable Checks

Include relevant controls to prevent omitted variable bias, ensuring that instruments are not correlated with unobserved factors.

Invalid IVs

Additional Best Practices

Leverage Intuition and Reduced-Form Regressions

Confirm that instruments behave as expected (e.g., litigation increases prison populations, which then reduce crime).

Avoid Data Mining

Pretesting and selecting instruments based on theoretical reasoning rather than fishing for significance reduces invalidity risk.

Caution with Over-Identification Tests

Recognize that failing to reject the null does not guarantee validity, especially if instruments share common rationales.

Weak IVs

Problems caused by them

Finite-sample bias

two-stage least squares (2SLS) estimates can be biased, sometimes more than ordinary least squares (OLS).

Underestimated standard errors

Leads to overstated statistical significance.

Distorted inference

Confidence intervals and hypothesis tests become unreliable, especially in over-identified models.

Weak IVs

Strategies to Address Weak Instruments

Stock-Yogo Test

Checks if instruments are strong enough to produce reliable estimates (e.g., bias less than 10% of OLS).

Conditional likelihood ratio tests (Moreira, Andrews)

Adjust for weak instruments, providing valid hypothesis testing and confidence intervals.

$$LRT = -2\ln\left(\frac{L_0}{L_1}\right)$$

Avoid relying solely on 2SLS

Always evaluate instrument strength before inference; avoid relying solely on 2SLS when instruments are weak.

Conclusion

Key Takeaways

The validity and strength of instruments are crucial for credible IV estimates. Weak or invalid IVs can lead to significant biases and unreliable inferences, especially when instruments are "almost valid" but weak.

Rigorous testing and validation are essential to avoid misleading results.

Final Remarks

Navigating Heterogeneity and Instrumental Variable Challenges

Addressing heterogeneity requires understanding which aspects of responses are economically relevant and choosing appropriate econometric techniques.

Recognizing that IV estimates may not reflect average effects across populations underscores the importance of careful instrument selection and interpretation. Diligence, theoretical grounding, and robustness checks are vital for credible causal inference.

Ending remarks

I am delighted for your attention.

Q & A?